

Summary

Over a decade experience in C and C++.

Education

- MS - Electrical Engineering New York University, 2009
- BS - Electrical Engineering & Computer Science University of California Berkeley, 2006

Career

- Software Verification Engineer - Contracts (Ontario) 04/2022 - Present

Developed functional, integrated and unit tests in C/C++ for:

- DO-178C verification of a safety-critical real-time operating system (RTOS),
- DO-178C verification of a Vulkan graphics API implementation for embedded GPUs, and
- pre-silicon verification of AR/VR firmware on Haps and Zebu FPGA emulation platforms.

- Staff System Design Engineer - AMD (Ontario) 03/2021 - 03/2022

Maintained, extended and developed Power Delivery and DisplayPort Alt Mode firmware for USB Type C controllers on Radeon RX 6000 graphic cards.

- Test Engineer - Creation Technologies (Ontario) 05/2020 - 02/2021

Oversaw system-level hardware and software verification of defense-sector RF products including wireless jammers and reconnaissance drones.

- Sr System Verification Engineer - Peraso (Ontario) 12/2018 - 03/2020

Developed system-level functional tests in JavaScript for the verification of PHY and MAC firmware from 60GHz mmWave RF modules.

- Product Engineer - Tallysman (Ontario) 01/2018 - 12/2018

Developed a C++ test framework with Microsoft SQL Server backend for an automated electromechanical test platform with an anechoic chamber for antenna characterization.

- Sr Test Engineer - Skyworks (California) 03/2014 - 05/2016

Developed C test solutions on in-house custom ATEs for electrical verification of high-volume GSM, UMTS and LTE cellular RF power amplifiers.

- Sr Test Engineer - Semtech (California) 09/2010 - 03/2014
Developed C++ test suites on V93000 ATEs for electrical verification of 40G/100G optical transceiver serializer-deserializer pairs.
- Electrical Engineer - LH Tag and Label (California) 07/2009 - 06/2010
Designed PCBs and developed C firmware for bare-metal embedded systems with motor, temperature and display controllers from textile label printing and cutting machines.

Skills

- Languages: C, C++, Python, Matlab, SQL, JavaScript
- Build Tools: GCC, Clang, Visual Studio, Make, Buck2
- Debugger: Lauterbach TRACE32
- Source Control: Git/GitHub, Mercurial, Perforce, SVN
- Requirement/Change Control: IBM DOORS, IBM RTC, Codebeamer
- Continuous Integration: TeamCity
- Agile: Jira
- Interfaces: USB, GPIB, UART/RS232, I2C, SPI, MIPI